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(54) **FAN ASSEMBLY**

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Mar. 1, 2010, now Pat. No. 7,972,111.

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F04D 29/54 (2006.01)

(52) **U.S. Cl.**
USPC **415/211.2**

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415/206; 416/170 R; 417/423.1, 423.14,
417/423.15; 310/67 R, 71; 174/68.1, 68.3,
174/72 A, 97, 101

See application file for complete search history.

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Primary Examiner — Edward Look

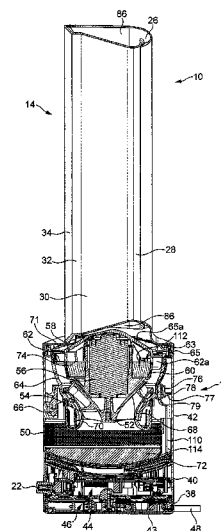
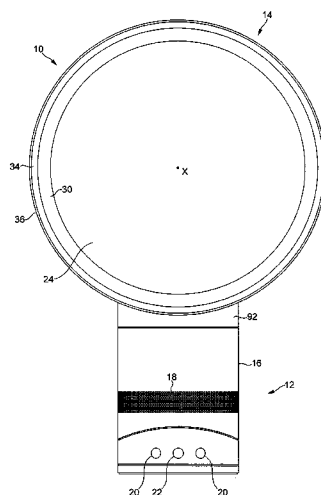
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(57) **ABSTRACT**

A fan assembly for creating an air current includes a nozzle mounted on a base. The base comprises an outer casing, an impeller housing located within the outer casing, the impeller housing having an air inlet and an air outlet, an impeller located within the impeller housing and a motor for driving the impeller to create an air flow through the impeller housing. The nozzle includes an interior passage for receiving the air flow from the air outlet of the impeller housing and a mouth through which the air flow is emitted from the fan assembly.

20 Claims, 10 Drawing Sheets



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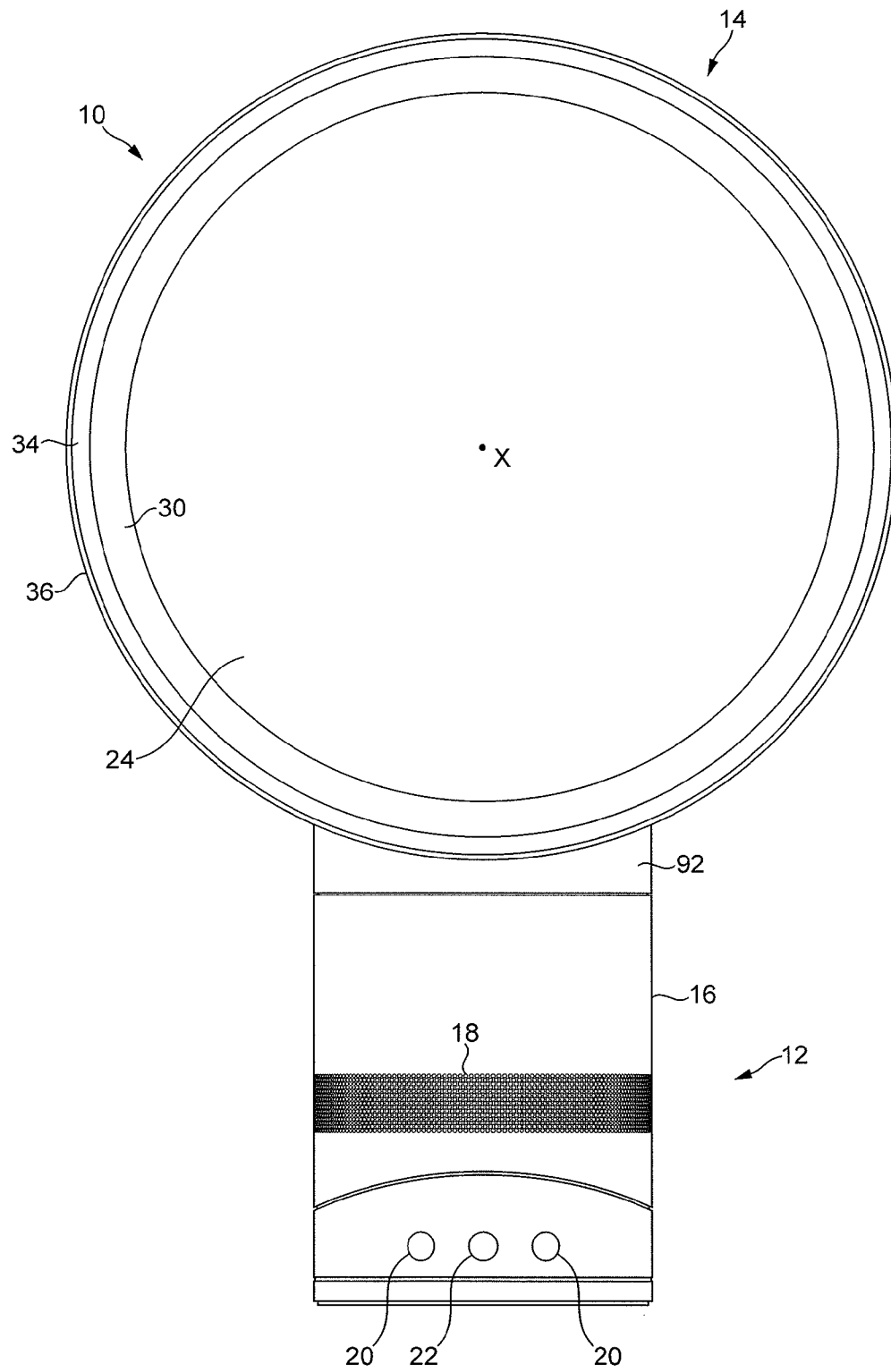


FIG. 1

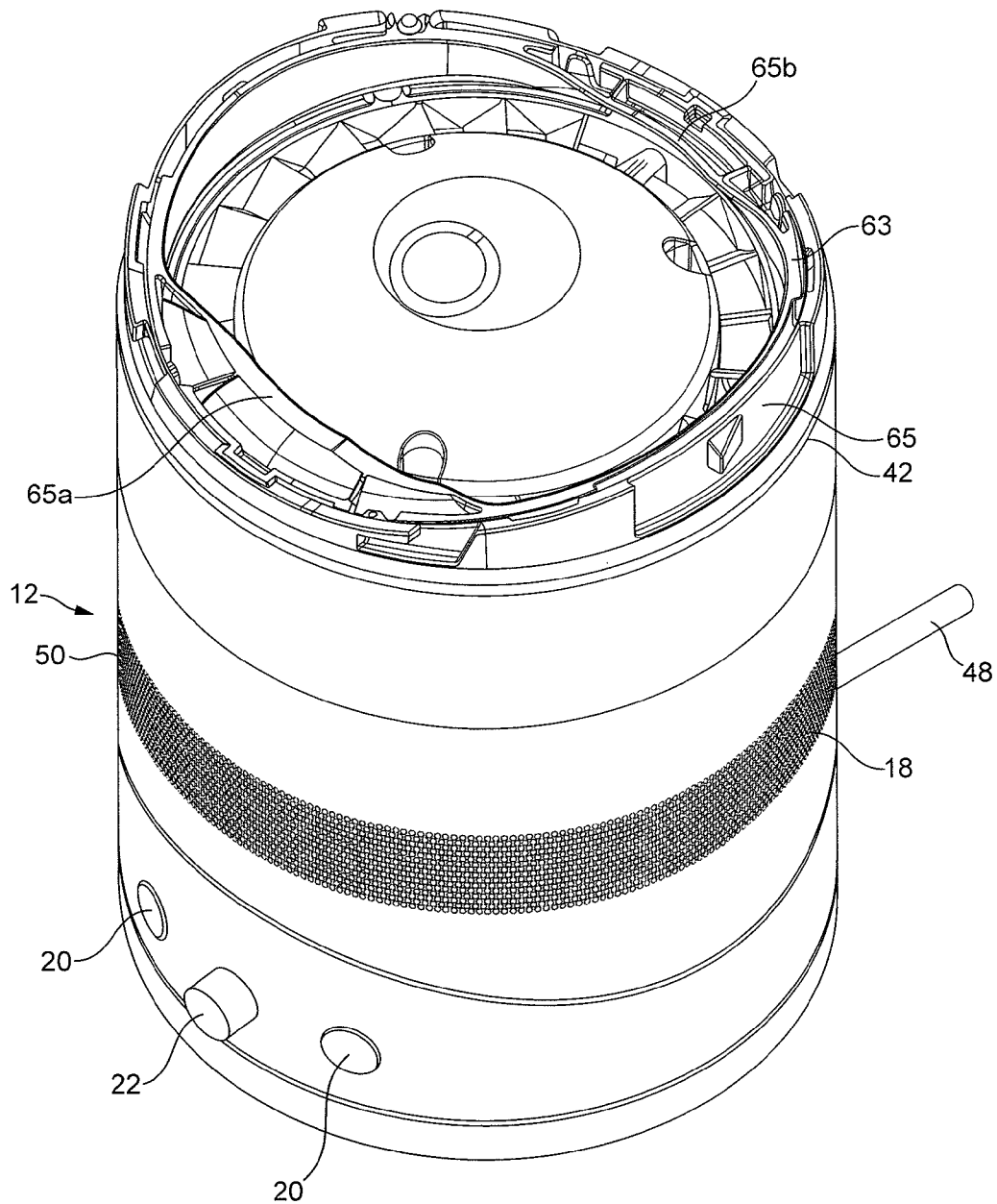


FIG. 2a

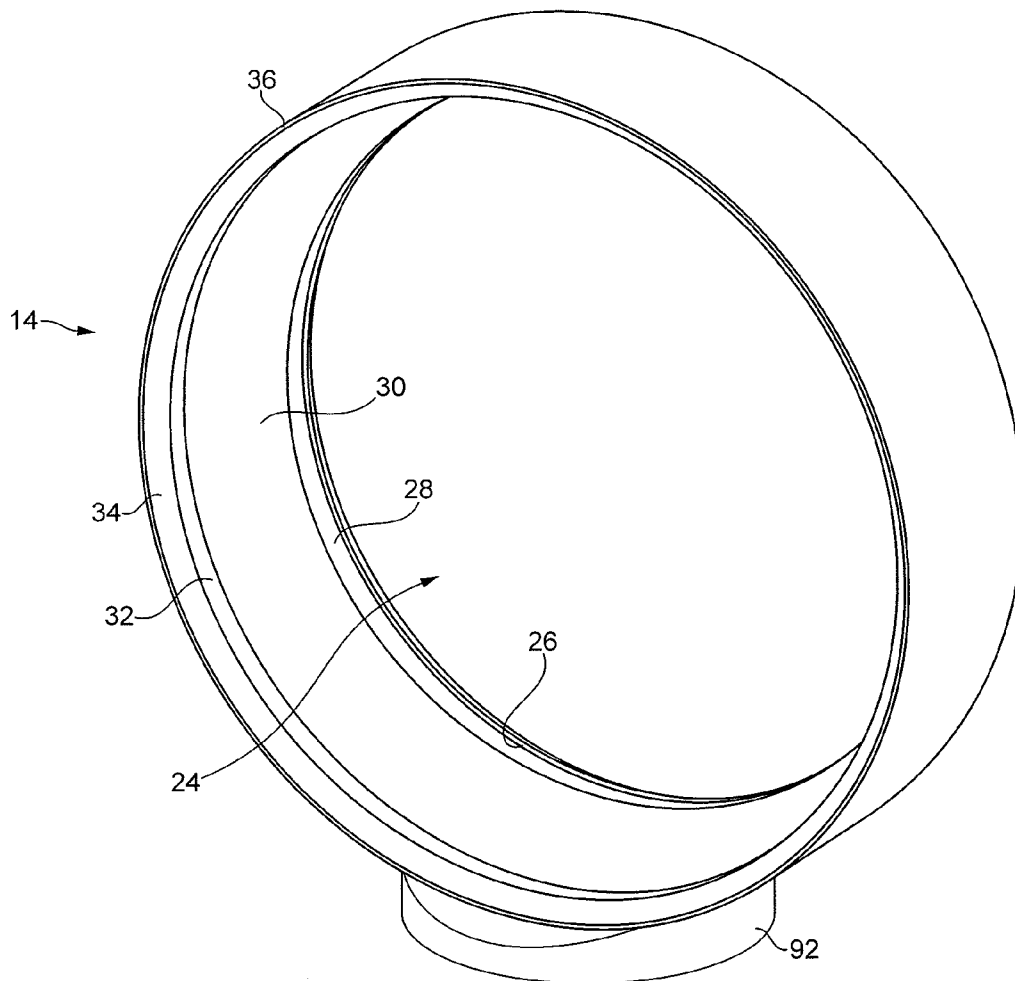


FIG. 2b

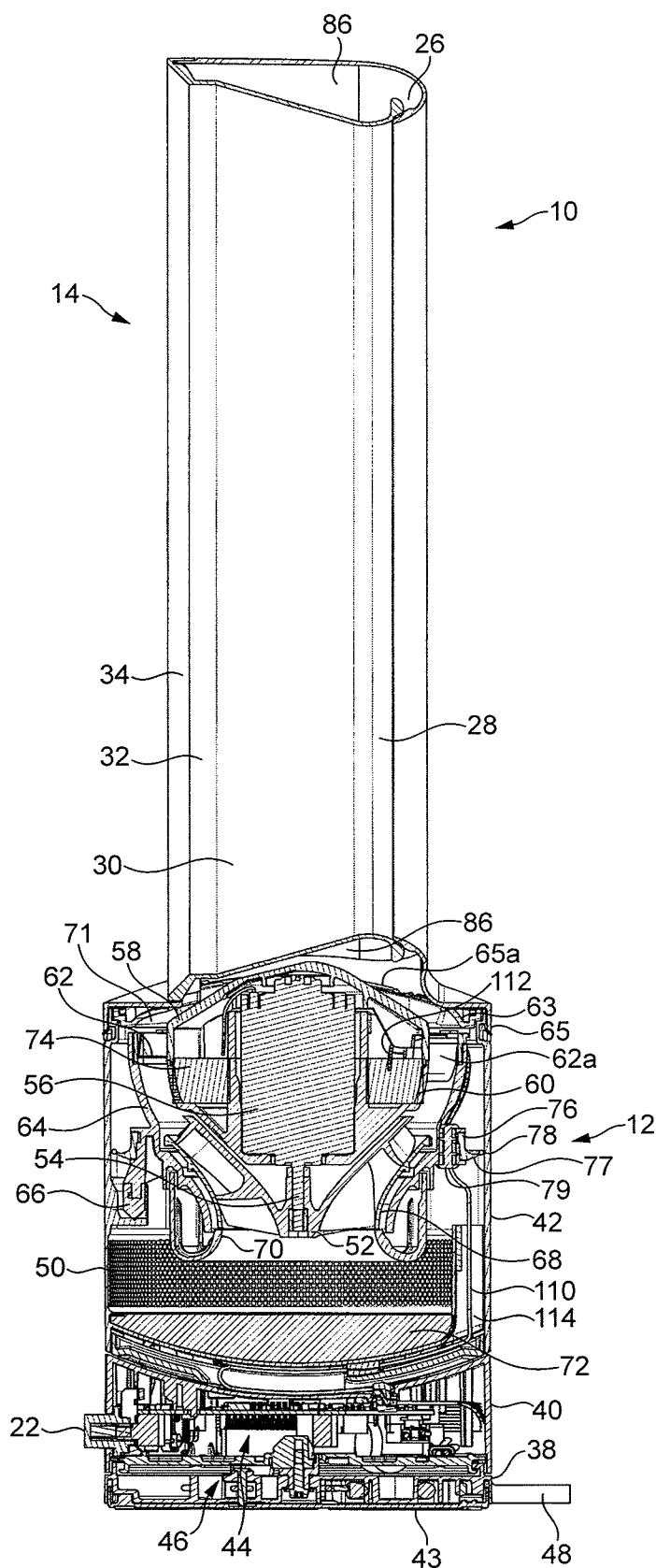


FIG. 3

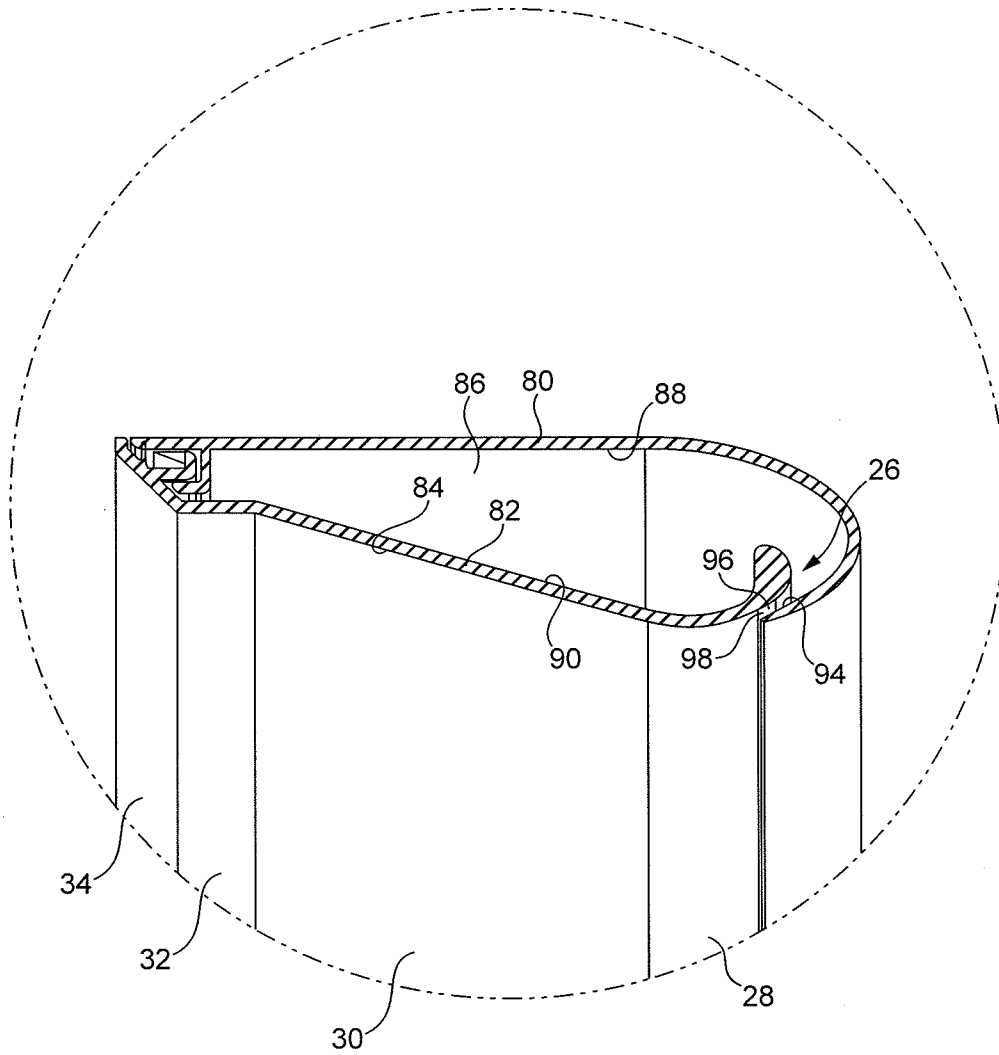


FIG. 4

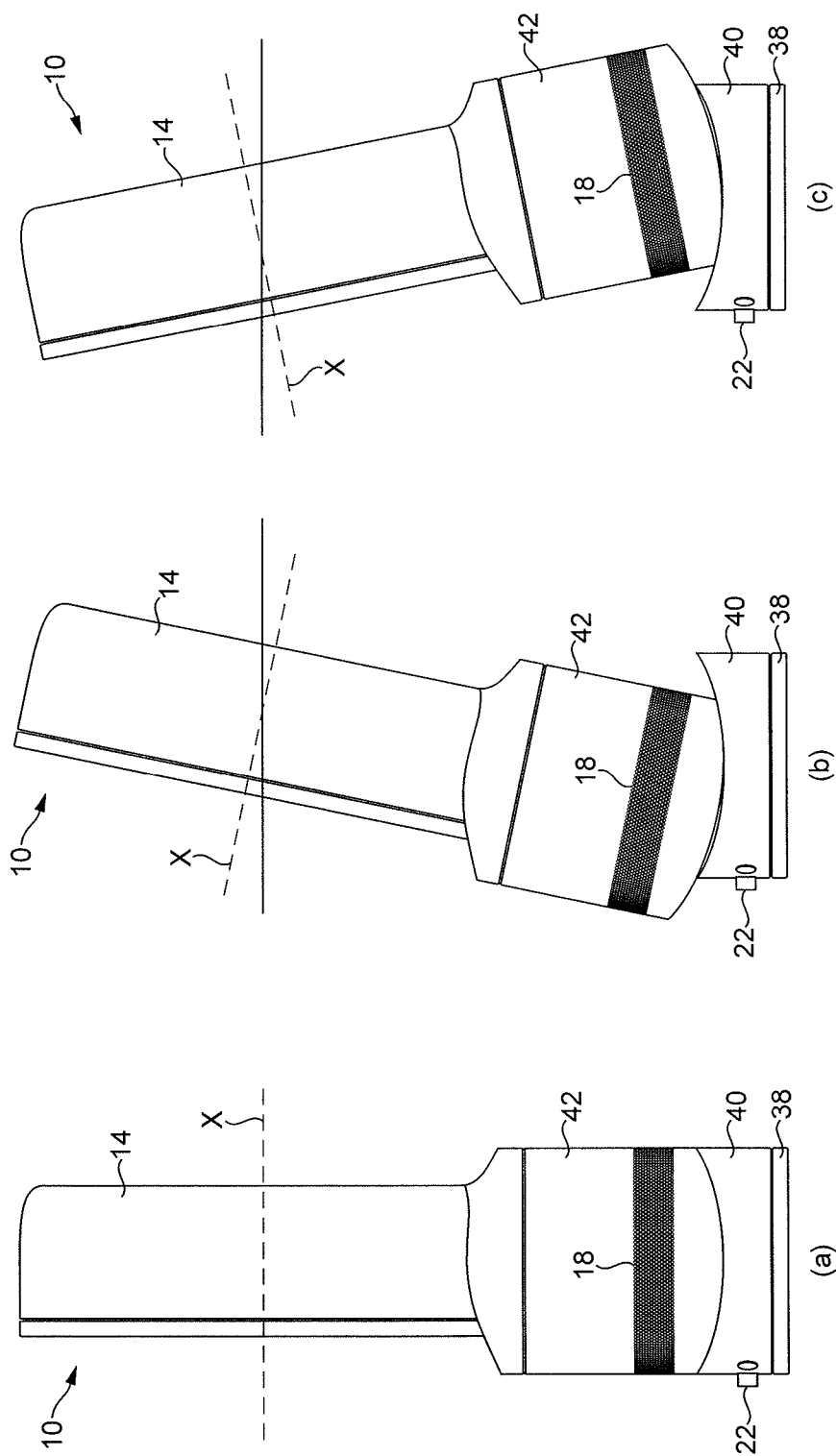


FIG. 5

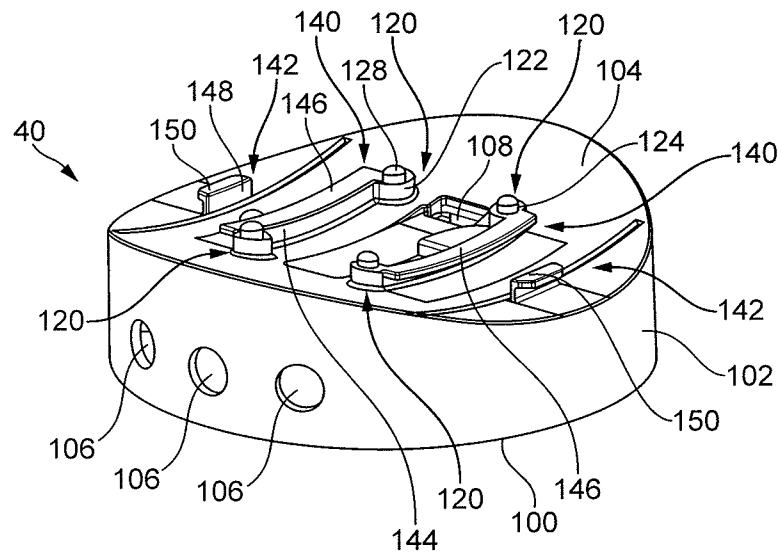


FIG. 6

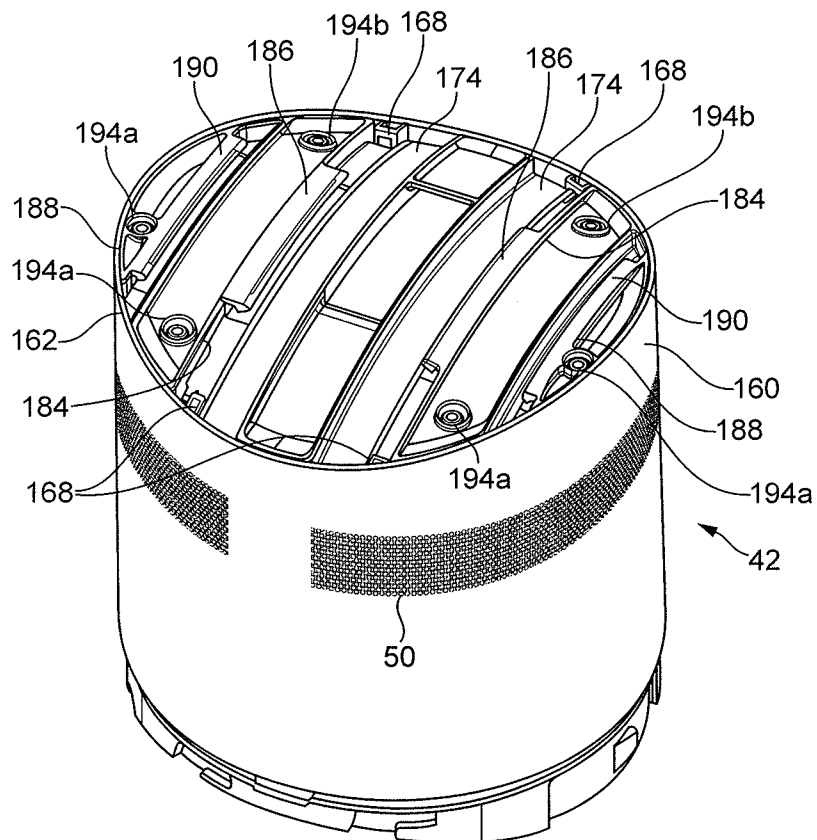


FIG. 7

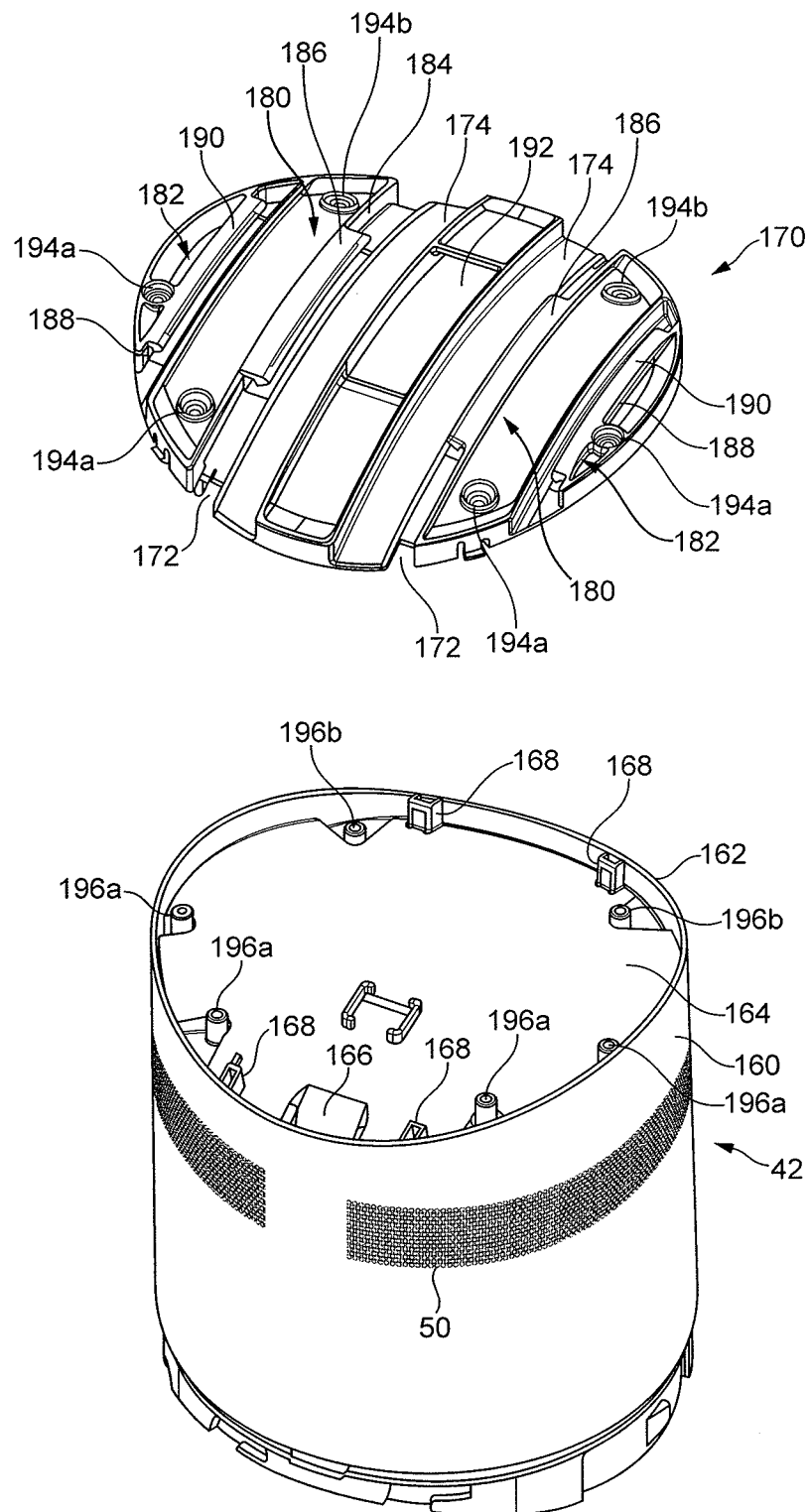


FIG. 8

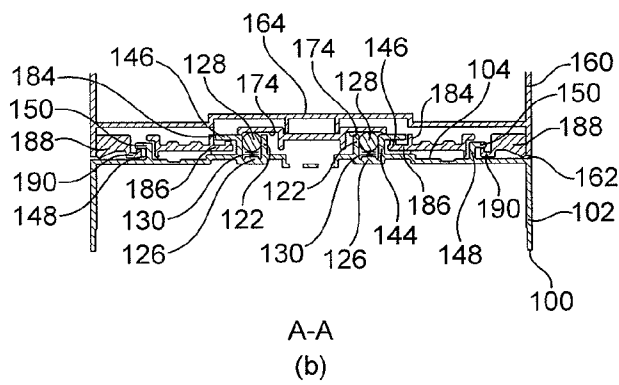
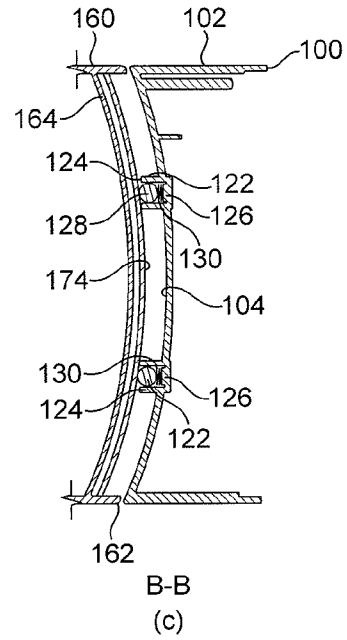
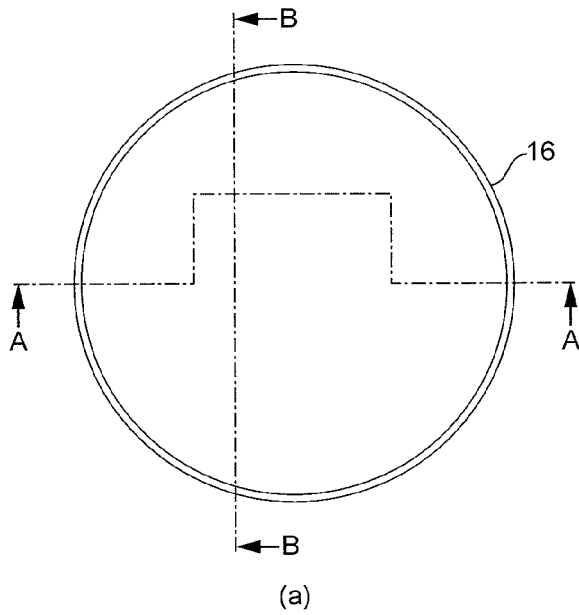
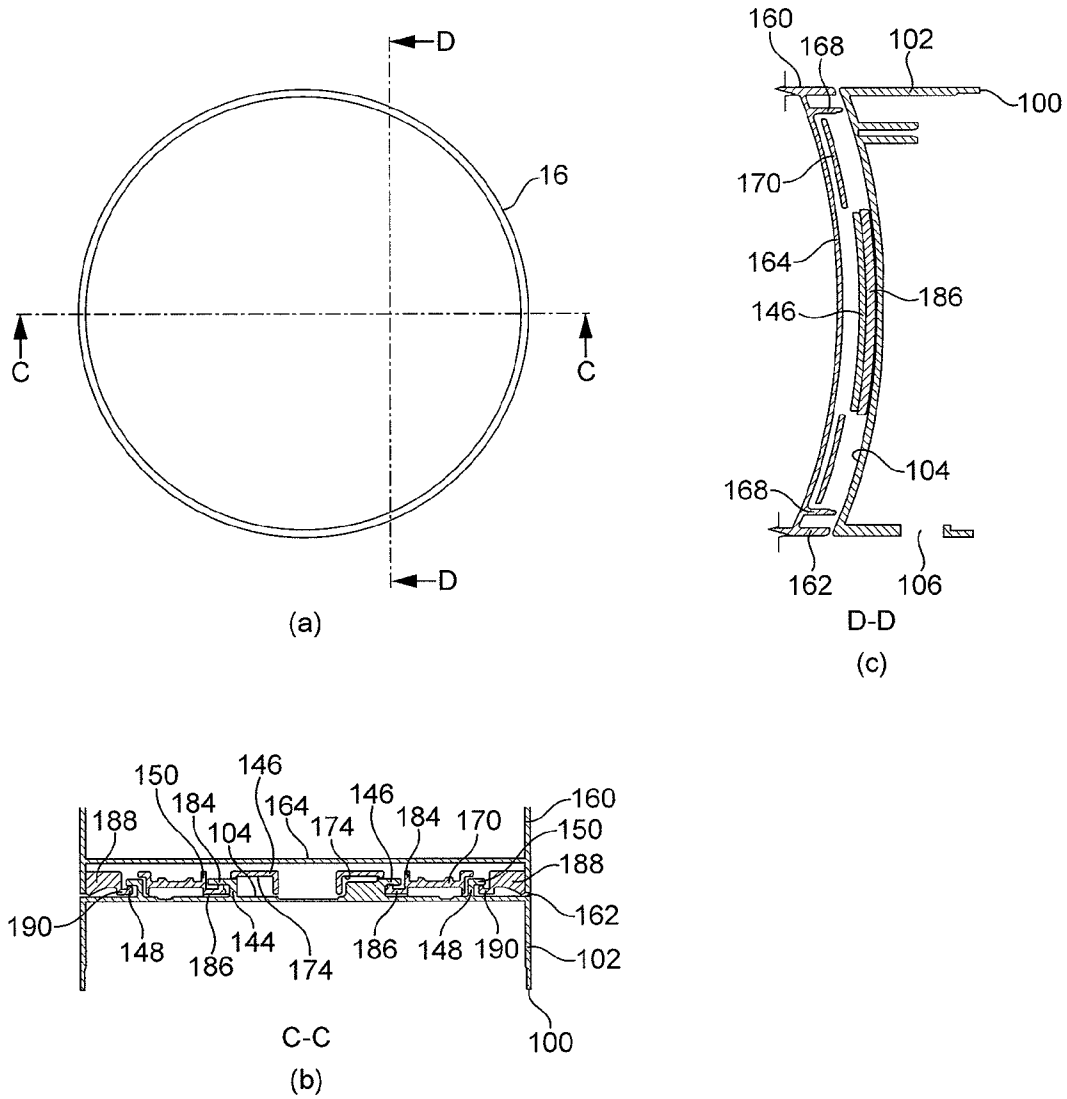


FIG. 9



1

FAN ASSEMBLY

REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 13/114,695, filed May 24, 2011, which is a continuation of U.S. patent application Ser. No. 12/715,076, filed Mar. 1, 2010, now U.S. Pat. No. 7,972,111, which claims the priority of United Kingdom Application No. 0903695.5 filed Mar. 4, 2009, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to a fan assembly. Particularly, but not exclusively, the present invention relates to a domestic fan, such as a desk fan, for creating air circulation and air current in a room, in an office or other domestic environment.

BACKGROUND OF THE INVENTION

A conventional domestic fan typically includes a set of blades or vanes mounted for rotation about an axis, and drive apparatus for rotating the set of blades to generate an air flow. The movement and circulation of the air flow creates a 'wind chill' or breeze and, as a result, the user experiences a cooling effect as heat is dissipated through convection and evaporation.

Such fans are available in a variety of sizes and shapes. For example, a ceiling fan can be at least 1 m in diameter, and is usually mounted in a suspended manner from the ceiling to provide a downward flow of air to cool a room. On the other hand, desk fans are often around 30 cm in diameter, and are usually free standing and portable. Other types of fan can be attached to the floor or mounted on a wall. Fans such as that disclosed in U.S. D 103,476 and U.S. Pat. No. 1,767,060 are suitable for standing on a desk or a table.

A disadvantage of this type of fan is that the air flow produced by the rotating blades of the fan is generally not uniform. This is due to variations across the blade surface or across the outward facing surface of the fan. The extent of these variations can vary from product to product and even from one individual fan machine to another. These variations result in the generation of an uneven or 'choppy' air flow which can be felt as a series of pulses of air and which can be uncomfortable for a user. In addition, this type of fan can be noisy and the noise generated may become intrusive with prolonged use in a domestic environment. A further disadvantage is that the cooling effect created by the fan diminishes with distance from the user. This means that the fan must be placed in close proximity to the user in order for the user to experience the cooling effect of the fan.

An oscillating mechanism may be employed to rotate the outlet from the fan so that the air flow is swept over a wide area of a room. In this way the direction of air flow from the fan can be altered. In addition the drive apparatus may rotate the set of blades at a variety of speeds to optimise the airflow output by the fan. The blade speed adjustment and oscillating mechanism can lead to some improvement in the quality and uniformity of the air flow felt by a user although the characteristic 'choppy' air flow remains.

Some fans, sometimes known as air circulators, generate a cooling flow of air without the use of rotating blades. Fans such as those described in U.S. Pat. No. 2,488,467 and JP 56-167897 have large base body portions including a motor and an impeller for generating an air flow in the base body.

2

The air flow is channeled from the base body to an air discharge slot from which the air flow is projected forward towards a user. The fan of U.S. Pat. No. 2,488,467 emits air flow from a series of concentric slots, whereas the fan of JP 56-167897 channels the air flow to a neck piece leading to a single air discharging slot.

A fan that attempts to provide cooling air flow through a slot without the use of rotating blades requires an efficient transfer of air flow from the base body to the slot. The air flow is constricted as it is channeled into the slot and this constriction creates pressure in the fan which must be overcome by the air flow generated by the motor and the impeller in order to project the air flow from the slot. Any inefficiencies in the system, for example losses through the fan housing, will reduce the air flow from the fan. The high efficiency requirement restricts the options for the use of motors and other means for creating air flow. This type of fan can be noisy as vibrations generated by the motor and impeller tend to be transmitted and amplified.

SUMMARY OF THE INVENTION

The present invention provides a fan assembly for creating an air current, the fan assembly comprising a nozzle mounted on a base comprising an outer casing, an impeller housing located within the outer casing, the impeller housing having an air inlet and an air outlet, an impeller located within the impeller housing and a motor for driving the impeller to create an air flow through the impeller housing, the nozzle comprising an interior passage for receiving the air flow from the air outlet of the impeller housing and a mouth through which the air flow is emitted from the fan assembly, wherein a flexible sealing member is located between the outer casing and the impeller housing.

The flexible sealing member inhibits the return of air to the air inlet along a path extending between the outer casing and the impeller housing, forcing the pressurized air flow generated by the impeller to be output through the impeller housing and into the nozzle. With this fan assembly a substantially constant pressure difference can be maintained between the motor and the impeller in the base, including the air outlet of the impeller housing, and the air inlet and impeller housing. Without the flexible sealing member the efficiency of the fan assembly would be degraded due to fluctuating losses within the base. Advantageously, the flexible sealing member absorbs some vibration and noise from the motor that would otherwise be transmitted and amplified through the fan assembly by a rigid sealing member.

Preferably the flexible sealing member is connected to the impeller housing for ease of assembly and to improve the sealing function of the sealing member with the impeller housing. More preferably, the flexible sealing member is biased against the outer casing, and can provide an air-tight seal between the outer housing and the impeller housing. In a preferred embodiment a portion of the flexible sealing member remote from the impeller housing is biased against the outer casing to form a lip seal. The seal can prevent high pressure air flow generated by the impeller mixing with air at, or close to, atmospheric air pressure.

Preferably the base is substantially cylindrical. This arrangement can be compact with base dimensions that are small compared to those of the nozzle and compared to the size of the overall fan assembly. Advantageously, the invention can provide a fan assembly delivering a suitable cooling effect from a footprint smaller than that of prior art fans.

In a preferred embodiment the flexible sealing member comprises an annular sealing member surrounding the impeller housing.

ler housing. Preferably the flexible sealing member comprises a guide portion for guiding a cable to the motor. Advantageously, the inclusion of a guide portion in the sealing member, preferably in the form of a flexible collar, allows cabling, such as a power cable, to pass through the flexible sealing member while maintaining the separation of the atmospheric pressure and higher pressure air flow regions of the fan assembly. This arrangement can reduce noise generation within the fan and the motor.

Preferably there is a diffuser located within the impeller housing and downstream from the impeller. The impeller is preferably a mixed flow impeller. The motor is preferably a DC brushless motor to avoid frictional losses and carbon debris from the brushes used in a traditional brushed motor. Reducing carbon debris and emissions is advantageous in a clean or pollutant sensitive environment such as a hospital or around those with allergies. While induction motors, which are generally used in fans, also have no brushes, a DC brushless motor can provide a much wider range of operating speeds than an induction motor. In a preferred embodiment a power cable is connected to the motor through the diffuser. The diffuser preferably comprises a plurality of fins, with the power cable passing through one of said plurality of fins. Advantageously, this arrangement can enable the power cable to be incorporated into the components of the base, reducing the overall part count and the number of components and connections required in the base. Passing the power cable, preferably a ribbon cable, through one of the fins of the diffuser is a neat, compact solution for power connection to the motor.

The base of the fan assembly preferably comprises means for directing a portion of the air flow from the air outlet of the impeller housing towards the interior passage of the nozzle.

The direction in which air is emitted from the air outlet of the impeller housing is preferably substantially at a right angle to the direction in which the air flow passes through at least part of the interior passage. The interior passage is preferably annular, and is preferably shaped to divide the air flow into two air streams which flow in opposite directions around the opening. In the preferred embodiment, the air flow passes into at least part of the interior passage in a sideways direction, and the air is emitted from the air outlet of the impeller housing in a forward direction. In view of this, the means for directing a portion of the air flow from the air outlet of the impeller housing preferably comprises at least one curved vane. The or each curved vane is preferably shaped to change the direction of the air flow by around 90°. The curved vanes are shaped so that there is no significant loss in the velocity of the portions of the air flow as they are directed into the interior passage.

The fan assembly is preferably in the form of a bladeless fan assembly. Through use of a bladeless fan assembly an air current can be generated without the use of a bladed fan. Without the use of a bladed fan to project the air current from the fan assembly, a relatively uniform air current can be generated and guided into a room or towards a user. The air current can travel efficiently out from the outlet, losing little energy and velocity to turbulence.

The term 'bladeless' is used to describe a fan assembly in which air flow is emitted or projected forward from the fan assembly without the use of moving blades. Consequently, a bladeless fan assembly can be considered to have an output area, or emission zone, absent moving blades from which the air flow is directed towards a user or into a room. The output area of the bladeless fan assembly may be supplied with a primary air flow generated by one of a variety of different sources, such as pumps, generators, motors or other fluid

transfer devices, and which may include a rotating device such as a motor rotor and/or a bladed impeller for generating the air flow. The generated primary air flow can pass from the room space or other environment outside the fan assembly into the fan assembly, and then back out to the room space through the outlet.

Hence, the description of a fan assembly as bladeless is not intended to extend to the description of the power source and components such as motors that are required for secondary fan functions. Examples of secondary fan functions can include lighting, adjustment and oscillation of the fan assembly.

The base preferably comprises control means for controlling the fan assembly. For safety reasons and ease of use, it can be advantageous to locate control elements away from the nozzle so that the control functions, such as, for example, oscillation, tilting, lighting or activation of a speed setting, are not activated during a fan operation.

Preferably, the nozzle extends about an axis to define the opening through which air from outside the fan assembly is drawn by the air flow emitted from the mouth. Preferably, the nozzle surrounds the opening. The nozzle may be an annular nozzle which preferably has a height in the range from 200 to 600 mm, more preferably in the range from 250 to 500 mm. The base preferably comprises at least one air inlet through which air is drawn into the fan assembly by the impeller. Preferably, said at least one air inlet is arranged substantially orthogonal to said axis. This can provide a short, compact air flow path that minimises noise and frictional losses.

Preferably, the mouth of the nozzle extends about the opening, and is preferably annular. Preferably the nozzle extends about the opening by a distance in the range from 50 to 250 cm. The nozzle preferably comprises at least one wall defining the interior passage and the mouth, and wherein said at least one wall comprises opposing surfaces defining the mouth. Preferably, the mouth has an outlet, and the spacing between the opposing surfaces at the outlet of the mouth is in the range from 0.5 mm to 5 mm, more preferably in the range from 0.5 mm to 1.5 mm. The nozzle may preferably comprise an inner casing section and an outer casing section which define the mouth of the nozzle. Each section is preferably formed from a respective annular member, but each section may be provided by a plurality of members connected together or otherwise assembled to form that section. The outer casing section is preferably shaped so as to partially overlap the inner casing section. This can enable an outlet of the mouth to be defined between overlapping portions of the external surface of the inner casing section and the internal surface of the outer casing section of the nozzle. The nozzle may comprise a plurality of spacers for urging apart the overlapping portions of the inner casing section and the outer casing section of the nozzle. This can assist in maintaining a substantially uniform outlet width about the opening. The spacers are preferably evenly spaced along the outlet.

The maximum air flow of the air current generated by the fan assembly is preferably in the range from 300 to 800 liters per second, more preferably in the range from 500 to 800 liters per second.

The nozzle may comprise a Coanda surface located adjacent the mouth and over which the mouth is arranged to direct the air flow emitted therefrom. Preferably, the external surface of the inner casing section of the nozzle is shaped to define the Coanda surface. The Coanda surface preferably extends about the opening. A Coanda surface is a known type of surface over which fluid flow exiting an output orifice close to the surface exhibits the Coanda effect. The fluid tends to flow over the surface closely, almost 'clinging to' or 'hug-

5

ging' the surface. The Coanda effect is already a proven, well documented method of entrainment in which a primary air flow is directed over a Coanda surface. A description of the features of a Coanda surface, and the effect of fluid flow over a Coanda surface, can be found in articles such as Reba, Scientific American, Volume 214, June 1966 pages 84 to 92. Through use of a Coanda surface, an increased amount of air from outside the fan assembly is drawn through the opening by the air emitted from the mouth.

Preferably, an air flow enters the nozzle of the fan assembly from the base. In the following description this air flow will be referred to as primary air flow. The primary air flow is emitted from the mouth of the nozzle and preferably passes over a Coanda surface. The primary air flow entrains air surrounding the mouth of the nozzle, which acts as an air amplifier to supply both the primary air flow and the entrained air to the user. The entrained air will be referred to here as a secondary air flow. The secondary air flow is drawn from the room space, region or external environment surrounding the mouth of the nozzle and, by displacement, from other regions around the fan assembly, and passes predominantly through the opening defined by the nozzle. The primary air flow directed over the Coanda surface combined with the entrained secondary air flow equates to a total air flow emitted or projected forward from the opening defined by the nozzle. Preferably, the entrainment of air surrounding the mouth of the nozzle is such that the primary air flow is amplified by at least five times, more preferably by at least ten times, while a smooth overall output is maintained.

Preferably, the nozzle comprises a diffuser surface located downstream of the Coanda surface. The external surface of the inner casing section of the nozzle is preferably shaped to define the diffuser surface.

BRIEF DESCRIPTION OF THE DRAWINGS

An embodiment of the invention will now be described with reference to the accompanying drawings, in which:

FIG. 1 is a front view of a fan assembly;

FIG. 2(a) is a perspective view of the base of the fan assembly of FIG. 1;

FIG. 2(b) is a perspective view of the nozzle of the fan assembly of FIG. 1;

FIG. 3 is a sectional view through the fan assembly of FIG. 1;

FIG. 4 is an enlarged view of part of FIG. 3;

FIG. 5(a) is a side view of the fan assembly of FIG. 1 showing the fan assembly in an untilted position;

FIG. 5(b) is a side view of the fan assembly of FIG. 1 showing the fan assembly in a first tilted position;

FIG. 5(c) is a side view of the fan assembly of FIG. 1 showing the fan assembly in a second, tilted position;

FIG. 6 is a top perspective view of the upper base member of the fan assembly of FIG. 1;

FIG. 7 is a rear perspective view of the main body of the fan assembly of FIG. 1;

FIG. 8 is an exploded view of the main body of FIG. 7;

FIG. 9(a) illustrates the paths of two sectional views through the base when the fan assembly is in an untilted position;

FIG. 9(b) is a sectional view along line A-A of FIG. 9(a);

FIG. 9(c) is a sectional view along line B-B of FIG. 9(a);

FIG. 10(a) illustrates the paths of two further sectional views through the base when the fan assembly is in an untilted position;

FIG. 10(b) is a sectional view along line C-C of FIG. 10(a); and

FIG. 10(c) is a sectional view along line D-D of FIG. 10(a).

6

DETAILED DESCRIPTION OF THE INVENTION

FIG. 1 is a front view of a fan assembly 10. The fan assembly 10 is preferably in the form of a bladeless fan assembly comprising a base 12 and a nozzle 14 mounted on and supported by the base 12. With reference to FIG. 2(a), the base 12 comprises a substantially cylindrical outer casing 16 having a plurality of air inlets 18 in the form of apertures located in the outer casing 16 and through which a primary air flow is drawn into the base 12 from the external environment. The base 12 further comprises a plurality of user-operable buttons 20 and a user-operable dial 22 for controlling the operation of the fan assembly 10. In this example the base 12 has a height in the range from 200 to 300 mm, and the outer casing 16 has an external diameter in the range from 100 to 200 mm.

With reference also to FIG. 2(b), the nozzle 14 has an annular shape and defines a central opening 24. The nozzle 14 has a height in the range from 200 to 400 mm. The nozzle 14 comprises a mouth 26 located towards the rear of the fan assembly 10 for emitting air from the fan assembly 10 and through the opening 24. The mouth 26 extends at least partially about the opening 24. The inner periphery of the nozzle 14 comprises a Coanda surface 28 located adjacent the mouth 26 and over which the mouth 26 directs the air emitted from the fan assembly 10, a diffuser surface 30 located downstream of the Coanda surface 28 and a guide surface 32 located downstream of the diffuser surface 30. The diffuser surface 30 is arranged to taper away from the central axis X of the opening 24 in such a way so as to assist the flow of air emitted from the fan assembly 10. The angle subtended between the diffuser surface 30 and the central axis X of the opening 24 is in the range from 5 to 25°, and in this example is around 15°. The guide surface 32 is arranged at an angle to the diffuser surface 30 to further assist the efficient delivery of a cooling air flow from the fan assembly 10. The guide surface 32 is preferably arranged substantially parallel to the central axis X of the opening 24 to present a substantially flat and substantially smooth face to the air flow emitted from the mouth 26. A visually appealing tapered surface 34 is located downstream from the guide surface 32, terminating at a tip surface 36 lying substantially perpendicular to the central axis X of the opening 24. The angle subtended between the tapered surface 34 and the central axis X of the opening 24 is preferably around 45°. The overall depth of the nozzle 24 in a direction extending along the central axis X of the opening 24 is in the range from 100 to 150 mm, and in this example is around 110 mm.

FIG. 3 illustrates a sectional view through the fan assembly 10. The base 12 comprises a lower base member 38, an intermediary base member 40 mounted on the lower base member 38, and an upper base member 42 mounted on the intermediary base member 40. The lower base member 38 has a substantially flat bottom surface 43. The intermediary base member 40 houses a controller 44 for controlling the operation of the fan assembly 10 in response to depression of the user operable buttons 20 shown in FIGS. 1 and 2, and/or manipulation of the user operable dial 22. The intermediary base member 40 may also house an oscillating mechanism 46 for oscillating the intermediary base member 40 and the upper base member 42 relative to the lower base member 38. The range of each oscillation cycle of the upper base member 42 is preferably between 60° and 120°, and in this example is around 90°. In this example, the oscillating mechanism 46 is arranged to perform around 3 to 5 oscillation cycles per

minute. A mains power cable 48 extends through an aperture formed in the lower base member 38 for supplying electrical power to the fan assembly 10.

The upper base member 42 of the base 12 has an open upper end. The upper base member 42 comprises a cylindrical grille mesh 50 in which an array of apertures is formed. In between each aperture are side wall regions known as 'lands'. The apertures provide the air inlets 18 of the base 12. A percentage of the total surface area of the cylindrical base is an open area equivalent to the total surface area of the apertures. In the illustrated embodiment the open area is 33% of the total mesh area, each aperture has a diameter of 1.2 mm and 1.8 mm from aperture centre to aperture centre, providing 0.6 mm of land in between each aperture. Aperture open area is required for air flow into the fan assembly, but large apertures can transmit vibrations and noise from the motor to the external environment. An open area of around 30% to 45% provides a compromise between lands to inhibit the emission of noise and openings for free, unrestricted inflow of air into the fan assembly.

The upper base member 42 houses an impeller 52 for drawing the primary air flow through the apertures of the grille mesh 50 and into the base 12. Preferably, the impeller 52 is in the form of a mixed flow impeller. The impeller 52 is connected to a rotary shaft 54 extending outwardly from a motor 56. In this example, the motor 56 is a DC brushless motor having a speed which is variable by the controller 44 in response to user manipulation of the dial 22. The maximum speed of the motor 56 is preferably in the range from 5,000 to 10,000 rpm. The motor 56 is housed within a motor bucket comprising an upper portion 58 connected to a lower portion 60. The motor bucket is retained within the upper base member 42 by a motor bucket retainer 63. The upper end of the upper base member 42 comprises a cylindrical outer surface 65. The motor bucket retainer 63 is connected to the open upper end of the upper base member 42, for example by a snap-fit connection. The motor 56 and its motor bucket are not rigidly connected to the motor bucket retainer 63, allowing some movement of the motor 56 within the upper base member 42.

The motor bucket retainer 63 comprises curved vane portions 65a and 65b extending inwardly from the upper end of the motor bucket retainer 63. Each curved vane 65a, 65b overlaps a part of the upper portion 58 of the motor bucket. Thus the motor bucket retainer 63 and the curved vanes 65a and 65b act to secure and hold the motor bucket in place during movement and handling. In particular, the motor bucket retainer 63 prevents the motor bucket becoming dislodged and falling towards the nozzle 14 if the fan assembly 10 becomes inverted.

One of the upper portion 58 and the lower portion 60 of the motor bucket comprises a diffuser 62 in the form of a stationary disc having spiral fins 62a, and which is located downstream from the impeller 52. One of the spiral fins 62a has a substantially inverted U-shaped cross-section when sectioned along a line passing vertically through the upper base member 42. This spiral fin 62a is shaped to enable a power connection cable to pass through the fin 62a.

The motor bucket is located within, and mounted on, an impeller housing 64. The impeller housing 64 is, in turn, mounted on a plurality of angularly spaced supports 66, in this example three supports, located within the upper base member 42 of the base 12. A generally frusto-conical shroud 68 is located within the impeller housing 64. The shroud 68 is shaped so that the outer edges of the impeller 52 are in close proximity to, but do not contact, the inner surface of the shroud 68. A substantially annular inlet member 70 is con-

nected to the bottom of the impeller housing 64 for guiding the primary air flow into the impeller housing 64. The top of the impeller housing 64 comprises a substantially annular air outlet 71 for guiding air flow emitted from the impeller housing 64. Preferably, the base 12 further comprises silencing foam for reducing noise emissions from the base 12. In this example, the upper base member 42 of the base 12 comprises a disc-shaped foam member 72 located towards the base of the upper base member 42, and a substantially annular foam member 74 located within the motor bucket.

A flexible sealing member is mounted on the impeller housing 64. The flexible sealing member inhibits the return of air to the air inlet member 70 along a path extending between the outer casing 16 and the impeller housing 64 by separating the primary air flow drawn in from the external environment from the air flow emitted from the air outlet 71 of the impeller 52 and diffuser 62. The sealing member preferably comprises a lip seal 76. The sealing member is annular in shape and surrounds the impeller housing 64, extending outwardly from the impeller housing 64 towards the outer casing 16. In the illustrated embodiment the diameter of the sealing member is greater than the radial distance from the impeller housing 64 to the outer casing 16. Thus the outer portion 77 of the sealing member is biased against the outer casing 16 and caused to extend along the inner face of the outer casing 16, forming a lip. The lip seal 76 of the preferred embodiment tapers and narrows to a tip 78 as it extends away from the impeller housing 64 and towards the outer casing 16. The lip seal 76 is preferably formed from rubber.

The lip seal 76 further comprises a guide portion for guiding a power connection cable to the motor 56. The guide portion 79 of the illustrated embodiment is formed in the shape of a collar and may be a grommet.

FIG. 4 illustrates a sectional view through the nozzle 14. The nozzle 14 comprises an annular outer casing section 80 connected to and extending about an annular inner casing section 82. Each of these sections may be formed from a plurality of connected parts, but in this embodiment each of the outer casing section 80 and the inner casing section 82 is formed from a respective, single moulded part. The inner casing section 82 defines the central opening 24 of the nozzle 14, and has an external peripheral surface 84 which is shaped to define the Coanda surface 28, diffuser surface 30, guide surface 32 and tapered surface 34.

The outer casing section 80 and the inner casing section 82 together define an annular interior passage 86 of the nozzle 14. Thus, the interior passage 86 extends about the opening 24. The interior passage 86 is bounded by the internal peripheral surface 88 of the outer casing section 80 and the internal peripheral surface 90 of the inner casing section 82. The outer casing section 80 comprises a base 92 which is connected to, and over, the open upper end of the upper base member 42 of the base 12, for example by a snap-fit connection. The base 92 of the outer casing section 80 comprises an aperture through which the primary air flow enters the interior passage 86 of the nozzle 14 from the upper end of the upper base member 42 of the base 12 and the open upper end of the motor bucket retainer 63.

The mouth 26 of the nozzle 14 is located towards the rear of the fan assembly 10. The mouth 26 is defined by overlapping, or facing, portions 94, 96 of the internal peripheral surface 88 of the outer casing section 80 and the external peripheral surface 84 of the inner casing section 82, respectively. In this example, the mouth 26 is substantially annular and, as illustrated in FIG. 4, has a substantially U-shaped cross-section when sectioned along a line passing diametrically through the nozzle 14. In this example, the overlapping portions 94, 96 of

the internal peripheral surface **88** of the outer casing section **80** and the external peripheral surface **84** of the inner casing section **82** are shaped so that the mouth **26** tapers towards an outlet **98** arranged to direct the primary flow over the Coanda surface **28**. The outlet **98** is in the form of an annular slot, preferably having a relatively constant width in the range from 0.5 to 5 mm. In this example the outlet **98** has a width of around 1.1 mm. Spacers may be spaced about the mouth **26** for urging apart the overlapping portions **94**, **96** of the internal peripheral surface **88** of the outer casing section **80** and the external peripheral surface **84** of the inner casing section **82** to maintain the width of the outlet **98** at the desired level. These spacers may be integral with either the internal peripheral surface **88** of the outer casing section **80** or the external peripheral surface **84** of the inner casing section **82**.

Turning now to FIGS. **5(a)**, **5(b)** and **5(c)**, the upper base member **42** is moveable relative to the intermediary base member **40** and the lower base member **38** of the base **12** between a first fully tilted position, as illustrated in FIG. **5(b)**, and a second fully tilted position, as illustrated in FIG. **5(c)**. This axis X is preferably inclined by an angle of around 10° as the main body is moved from an untilted position, as illustrated in FIG. **5(a)** to one of the two fully tilted positions. The outer surfaces of the upper base member **42** and the intermediary base member **40** are shaped so that adjoining portions of these outer surfaces of the upper base member **42** and the base **12** are substantially flush when the upper base member **42** is in the untilted position.

With reference to FIG. **6**, the intermediary base member **40** comprises an annular lower surface **100** which is mounted on the lower base member **38**, a substantially cylindrical side wall **102** and a curved upper surface **104**. The side wall **102** comprises a plurality of apertures **106**. The user-operable dial **22** protrudes through one of the apertures **106** whereas the user-operable buttons **20** are accessible through the other apertures **106**. The curved upper surface **104** of the intermediary base member **40** is concave in shape, and may be described as generally saddle-shaped. An aperture **108** is formed in the upper surface **104** of the intermediary base member **40** for receiving an electrical cable **110** (shown in FIG. **3**) extending from the motor **56**.

Returning to FIG. **3** the electrical cable **110** is a ribbon cable attached to the motor at joint **112**. The electrical cable **110** extending from the motor **56** passes out of the lower portion **60** of the motor bucket through spiral fin **62a**. The passage of the electrical cable **110** follows the shaping of the impeller housing **64** and the guide portion **79** of the lip seal **76** is shaped to enable the electrical cable **110** to pass through flexible sealing member. The collar of the lip seal **76** enables the electrical cable to be clamped and held within the upper base member **42**. A cuff **114** accommodates the electrical cable **110** within the lower portion of the upper base member **42**.

The intermediary base member **40** further comprises four support members **120** for supporting the upper base member **42** on the intermediary base member **40**. The support members **120** project upwardly from the upper surface **104** of the intermediary base member **40**, and are arranged such that they are substantially equidistant from each other, and substantially equidistant from the centre of the upper surface **104**. A first pair of the support members **120** is located along the line B-B indicated in FIG. **9(a)**, and a second pair of the support members **120** is parallel with the first pair of support members **120**. With reference also to FIGS. **9(b)** and **9(c)**, each support member **120** comprises a cylindrical outer wall **122**, an open upper end **124** and a closed lower end **126**. The outer wall **122** of the support member **120** surrounds a rolling element **128** in

the form of a ball bearing. The rolling element **128** preferably has a radius which is slightly smaller than the radius of the cylindrical outer wall **122** so that the rolling element **128** is retained by and moveable within the support member **120**. The rolling element **128** is urged away from the upper surface **104** of the intermediary base member **40** by a resilient element **130** located between the closed lower end **126** of the support member **120** and the rolling element **128** so that part of the rolling element **128** protrudes beyond the open upper end **124** of the support member **120**. In this embodiment, the resilient member **130** is in the form of a coiled spring.

Returning to FIG. **6**, the intermediary base member **40** also comprises a plurality of rails for retaining the upper base member **42** on the intermediary base member **40**. The rails **42** relative to the intermediary base member **40** so that there is substantially no twisting or rotation of the upper base member **42** relative to the intermediary base member **40** as it is moved from or to a tilted position. Each of the rails extends in a direction substantially parallel to the axis X. For example, one of the rails lies along line D-D indicated in FIG. **10(a)**. In this embodiment, the plurality of rails comprises a pair of relatively long, inner rails **140** located between a pair of relatively short, outer rails **142**. With reference also to FIGS. **9(b)** and **10(b)**, each of the inner rails **140** has a cross-section in the form of an inverted L-shape, and comprises a wall **144** which extends between a respective pair of the support members **120**, and which is connected to, and upstanding from, the upper surface **104** of the intermediary base member **40**. Each of the inner rails **140** further comprises a curved flange **146** which extends along the length of the wall **144**, and which protrudes orthogonally from the top of the wall **144** towards the adjacent outer guide rail **142**. Each of the outer rails **142** also has a cross-section in the form of an inverted L-shape, and comprises a wall **148** which is connected to, and upstanding from, the upper surface **52** of the intermediary base member **40** and a curved flange **150** which extends along the length of the wall **148**, and which protrudes orthogonally from the top of the wall **148** away from the adjacent inner guide rail **140**.

With reference now to FIGS. **7** and **8**, the upper base member **42** comprises a substantially cylindrical side wall **160**, an annular lower end **162** and a curved base **164** which is spaced from lower end **162** of the upper base member **42** to define a recess. The grille **50** is preferably integral with the side wall **160**. The side wall **160** of the upper base member **42** has substantially the same external diameter as the side wall **102** of the intermediary base member **40**. The base **164** is convex in shape, and may be described generally as having an inverted saddle-shape. An aperture **166** is formed in the base **164** for allowing the cable **110** to extend from base **164** of the upper base member **42** into the cuff **114**. Two pairs of stop members **168** extend upwardly (as illustrated in FIG. **8**) from the periphery of base **164**. Each pair of stop members **168** is located along a line extending in a direction substantially parallel to the axis X. For example, one of the pairs of stop members **168** is located along line D-D illustrated in FIG. **10(a)**.

A convex tilt plate **170** is connected to the base **164** of the upper base member **42**. The tilt plate **170** is located within the recess of the upper base member **42**, and has a curvature which is substantially the same as that of the base **164** of the upper base member **42**. Each of the stop members **168** protrudes through a respective one of a plurality of apertures **172** located about the periphery of the tilt plate **170**. The tilt plate **170** is shaped to define a pair of convex races **174** for engaging the rolling elements **128** of the intermediary base member **40**.

11

Each race **174** extends in a direction substantially parallel to the axis X, and is arranged to receive the rolling elements **128** of a respective pair of the support members **120**, as illustrated in FIG. 9(c).

The tilt plate **170** also comprises a plurality of runners, each of which is arranged to be located at least partially beneath a respective rail of the intermediary base member **40** and thus co-operate with that rail to retain the upper base member **42** on the intermediary base member **40** and to guide the movement of the upper base member **42** relative to the intermediary base member **40**. Thus, each of the runners extends in a direction substantially parallel to the axis X. For example, one of the runners lies along line D-D indicated in FIG. 10(a). In this embodiment, the plurality of runners comprises a pair of relatively long, inner runners **180** located between a pair of relatively short, outer runners **182**. With reference also to FIGS. 9(b) and 10(b), each of the inner runners **180** has a cross-section in the form of an inverted L-shape, and comprises a substantially vertical wall **184** and a curved flange **186** which protrudes orthogonally and inwardly from part of the top of the wall **184**. The curvature of the curved flange **186** of each inner runner **180** is substantially the same as the curvature of the curved flange **146** of each inner rail **140**. Each of the outer runners **182** also has a cross-section in the form of an inverted L-shape, and comprises a substantially vertical wall **188** and a curved flange **190** which extends along the length of the wall **188**, and which protrudes orthogonally and inwardly from the top of the wall **188**. Again, the curvature of the curved flange **190** of each outer runner **182** is substantially the same as the curvature of the curved flange **150** of each outer rail **142**. The tilt plate **170** further comprises an aperture **192** for receiving the electrical cable **110**.

To connect the upper base member **42** to the intermediary base member **40**, the tilt plate **170** is inverted from the orientation illustrated in FIGS. 7 and 8, and the races **174** of the tilt plate **170** located directly behind and in line with the support members **120** of the intermediary base member **40**. The electrical cable **110** extending through the aperture **166** of the upper base member **42** may be threaded through the apertures **108**, **192** in the tilt plate **170** and the intermediary base member **40** respectively for subsequent connection to the controller **44**, as illustrated in FIG. 3. The tilt plate **170** is then slid over the intermediary base member **40** so that the rolling elements **128** engage the races **174**, as illustrated in FIGS. 9(b) and 9(c), the curved flange **190** of each outer runner **182** is located beneath the curved flange **150** of a respective outer rail **142**, as illustrated in FIGS. 9(b) and 10(b), and the curved flange **186** of each inner runner **180** is located beneath the curved flange **146** of a respective inner rail **140**, as illustrated in FIGS. 9(b), 10(b) and 10(c).

With the tilt plate **170** positioned centrally on the intermediary base member **40**, the upper base member **42** is lowered on to the tilt plate **170** so that the stop members **168** are located within the apertures **172** of the tilt plate **170**, and the tilt plate **170** is housed within the recess of the upper base member **42**. The intermediary base member **40** and the upper base member **42** are then inverted, and the base member **40** displaced along the direction of the axis X to reveal a first plurality of apertures **194a** located on the tilt plate **170**. Each of these apertures **194a** is aligned with a tubular protrusion **196a** on the base **164** of the upper base member **42**. A self-tapping screw is screwed into each of the apertures **194a** to enter the underlying protrusion **196a**, thereby partially connecting the tilt plate **170** to the upper base member **42**. The intermediary base member **40** is then displaced in the reverse direction to reveal a second plurality of apertures **194b**

12

located on the tilt plate **170**. Each of these apertures **194b** is also aligned with a tubular protrusion **196b** on the base **164** of the upper base member **42**. A self-tapping screw is screwed into each of the apertures **194b** to enter the underlying protrusion **196b** to complete the connection of the tilt plate **170** to the upper base member **42**.

When the upper base member **42** is attached to the intermediary base member **40** and the bottom surface **43** of the lower base member **38** positioned on a support surface, the upper base member **42** is supported by the rolling elements **128** of the support members **120**. The resilient elements **130** of the support members **120** urge the rolling elements **128** away from the closed lower ends **126** of the support members **120** by a distance which is sufficient to inhibit scraping of the upper surfaces of the intermediary base member **40** when the upper base member **42** is tilted. For example, as illustrated in each of FIGS. 9(b), 9(c), 10(b) and 10(c) the lower end **162** of the upper base member **42** is urged away from the upper surface **104** of the intermediary base member **40** to prevent contact therebetween when the upper base member **42** is tilted. Furthermore, the action of the resilient elements **130** urges the concave upper surfaces of the curved flanges **186**, **190** of the runners against the convex lower surfaces of the curved flanges **146**, **150** of the rails.

To tilt the upper base member **42** relative to the intermediary base member **40**, the user slides the upper base member **42** in a direction parallel to the axis X to move the upper base member **42** towards one of the fully tilted positions illustrated in FIGS. 5(b) and 5(c), causing the rolling elements **128** move along the races **174**. Once the upper base member **42** is in the desired position, the user releases the upper base member **42**, which is retained in the desired position by frictional forces generated through the contact between the concave upper surfaces of the curved flanges **186**, **190** of the runners and the convex lower surfaces of the curved flanges **146**, **150** of the rails acting to resist the movement under gravity of the upper base member **42** towards the untilted position illustrated in FIG. 5(a). The fully tilted positions of the upper base member **42** are defined by the abutment of one of each pair of stop members **168** with a respective inner rail **140**.

To operate the fan assembly **10** the user depresses an appropriate one of the buttons **20** on the base **12**, in response to which the controller **44** activates the motor **56** to rotate the impeller **52**. The rotation of the impeller **52** causes a primary air flow to be drawn into the base **12** through the air inlets **18**. Depending on the speed of the motor **56**, the primary air flow may be between 20 and 30 liters per second. The primary air flow passes sequentially through the impeller housing **64**, the upper end of the upper base member **42** and open upper end of the motor bucket retainer **63** to enter the interior passage **86** of the nozzle **14**. The primary air flow emitted from the air outlet **71** is in a forward and upward direction. Within the nozzle **14**, the primary air flow is divided into two air streams which pass in opposite directions around the central opening **24** of the nozzle **14**. Part of the primary airflow entering the nozzle **14** in a sideways direction passes into the interior passage **86** in a sideways direction without significant guidance, another part of the primary airflow entering the nozzle **14** in a direction parallel to the X axis is guided by the curved vane **65a**, **65b** of the motor bucket retainer **63** to enable the air flow to pass into the interior passage **86** in a sideways direction. The vane **65a**, **65b** enables air flow to be directed away from a direction parallel to the X axis. As the air streams pass through the interior passage **86**, air enters the mouth **26** of the nozzle **14**. The air flow into the mouth **26** is preferably substantially even about the opening **24** of the nozzle **14**. Within each section of the mouth **26**, the flow direction of the portion of

13

the air stream is substantially reversed. The portion of the air stream is constricted by the tapering section of the mouth 26 and emitted through the outlet 98.

The primary air flow emitted from the mouth 26 is directed over the Coanda surface 28 of the nozzle 14, causing a secondary air flow to be generated by the entrainment of air from the external environment, specifically from the region around the outlet 98 of the mouth 26 and from around the rear of the nozzle 14. This secondary air flow passes through the central opening 24 of the nozzle 14, where it combines with the primary air flow to produce a total air flow, or air current, projected forward from the nozzle 14. Depending on the speed of the motor 56, the mass flow rate of the air current projected forward from the fan assembly 10 may be up to 400 liters per second, preferably up to 600 liters per second, and the maximum speed of the air current may be in the range from 2.5 to 4 m/s.

The even distribution of the primary air flow along the mouth 26 of the nozzle 14 ensures that the air flow passes evenly over the diffuser surface 30. The diffuser surface 30 causes the mean speed of the air flow to be reduced by moving the air flow through a region of controlled expansion. The relatively shallow angle of the diffuser surface 30 to the central axis X of the opening 24 allows the expansion of the air flow to occur gradually. A harsh or rapid divergence would otherwise cause the air flow to become disrupted, generating vortices in the expansion region. Such vortices can lead to an increase in turbulence and associated noise in the air flow which can be undesirable, particularly in a domestic product such as a fan. The air flow projected forwards beyond the diffuser surface 30 can tend to continue to diverge. The presence of the guide surface 32 extending substantially parallel to the central axis X of the opening 30 further converges the air flow. As a result, the air flow can travel efficiently out from the nozzle 14, enabling the air flow can be experienced rapidly at a distance of several metres from the fan assembly 10.

The invention is not limited to the detailed description given above. Variations will be apparent to the person skilled in the art.

For example, the motor bucket retainer and the sealing member may have a different size and/or shape to that described above and may be located in a different position within the fan assembly. The technique of creating an air tight seal with the sealing member may be different and may include additional elements such as glue or fixings. The sealing member, the guide portion, the vanes and the motor bucket retainer may be formed from any material with suitable strength and flexibility or rigidity, for example foam, plastics, metal or rubber. The movement of the upper base member 42 relative to the base may be motorised, and actuated by user through depression of one of the buttons 20.

The invention claimed is:

1. A fan assembly for creating an air current, the fan assembly comprising a nozzle mounted on a base, the base comprising an outer casing, the outer casing comprising a plurality of air inlets, a plurality of angularly spaced supports located within the outer casing, an impeller housing mounted on the angularly spaced supports, the impeller housing having an air inlet and an air outlet, an impeller located within the impeller housing, a motor located within the impeller housing for driving the impeller to create an air flow through the impeller housing, a diffuser located within the impeller housing and downstream of the impeller, and a flexible sealing member for inhibiting the flow of air from the air outlet of the

14

impeller housing to the air inlet of the impeller housing along a path extending between the impeller housing and the outer casing, the nozzle comprising an interior passage for receiving the air flow from the air outlet of the impeller housing and a mouth through which the air flow is emitted from the fan assembly, the nozzle defining an opening through which air from outside the fan assembly is drawn by the air flow emitted from the mouth.

2. The fan assembly of claim 1, wherein the flexible sealing member is mounted on the impeller housing.

3. The fan assembly of claim 1, wherein the flexible sealing member is biased against the outer casing.

4. The fan assembly of claim 1, wherein the base is substantially cylindrical.

5. The fan assembly of claim 1, wherein the flexible sealing member comprises an annular sealing member surrounding the impeller housing.

6. The fan assembly of claim 1, wherein the base of the fan assembly comprises a vane for directing a portion of the air flow from the air outlet of the impeller housing towards the interior passage of the nozzle.

7. The fan assembly of claim 6, wherein the vane is curved.

8. The fan assembly of claim 6, wherein the vane is shaped to change the direction of the air flow by around 90°.

9. The fan assembly of claim 1, wherein the fan assembly is bladeless.

10. The fan assembly of claim 1, wherein the nozzle comprises an annular inner casing section and an annular outer casing section which together define the interior passage and the mouth.

11. The fan assembly of claim 10, wherein the mouth comprises an outlet located between an external surface of the inner casing section and an internal surface of the outer casing section.

12. The fan assembly of claim 11, wherein the outlet is in the form of a slot.

13. The fan assembly of claim 11, wherein the outlet has a width in the range from 0.5 to 5 mm.

14. The fan assembly of claim 1, wherein the nozzle comprises a Coanda surface located adjacent the mouth and over which the mouth is arranged to direct the air flow.

15. The fan assembly of claim 14, wherein the nozzle comprises a diffuser located downstream of the Coanda surface.

16. The fan assembly of claim 1, wherein the diffuser comprises a plurality of fins, and wherein the plurality of fins is mounted on the impeller housing.

17. The fan assembly of claim 1, wherein the base comprises a lower base member, an intermediary base member mounted on the lower base member, and an upper base member mounted on the intermediary base member, wherein the nozzle is mounted on the upper base member, the upper base member comprising the outer casing of the base.

18. The fan assembly of claim 17, wherein the intermediary base member comprises a controller for controlling operation of the fan assembly, and wherein the base comprises a power cable extending from the controller to the motor and around the impeller housing.

19. The fan assembly of claim 17, comprising an oscillation mechanism for oscillating the intermediary base member relative to the lower base member.

20. The fan assembly of claim 17, wherein the upper base member is tiltable relative to the intermediary base member.

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